

Name of investigator: Alaa Al Saad

Date:10/25/2022

- describing what tools, you used

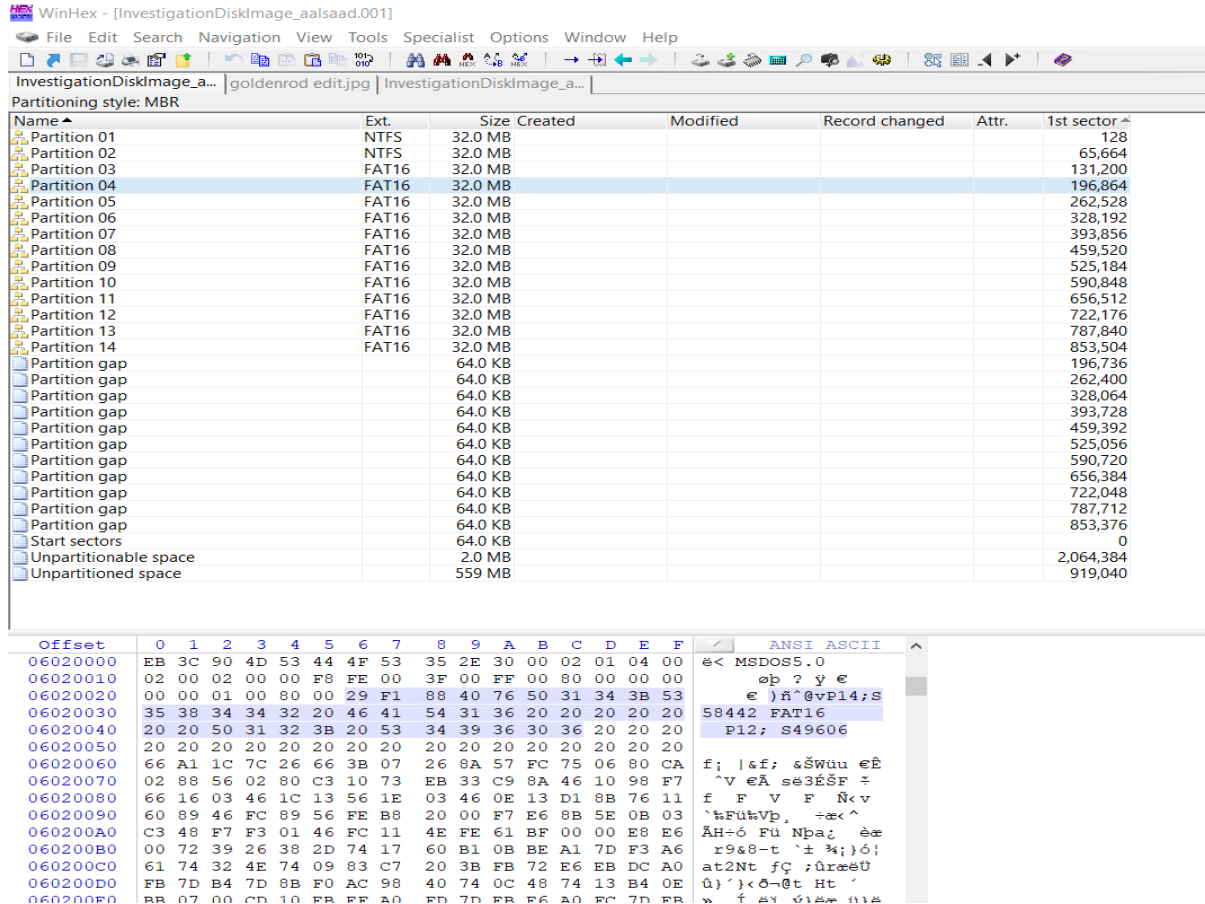


Figure 1: it shows screenshot of a forensic raw copy of the suspect's USB flash drive on Win Hex program.

Explanation: Win Hex tool which helps me to open inside it a forensic raw copy of the suspect's USB flash drive to search about clue on partitions four that I found two the partition numbers and logical partition sector addresses. I noted under ANSI ASCII two partitions (P14, S58442) and (P12,S49606) helping to find 8 account number.

case - Autopsy 4.19.3

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Data Sources

- InvestigatorDiskImage_aisaid.001_1 Host
 - vo15 [Unallocated: 0-127]
 - vo2 [NTFS / exFAT (DnD): 128-65663]
 - vo3 [NTFS / exFAT (DnD): 65664-131199]
 - vo4 [NTFS FAT 16 (DnD): 131200-360735]
 - vo7 [Unallocated: 360736-59663]
 - vo8 [NTFS FAT 16 (DnD): 59664-262399]
 - vo11 [Unallocated: 262400-362327]
 - vo12 [NTFS FAT 16 (DnD): 362328-328063]
 - vo15 [Unallocated: 328064-328191]
 - vo16 [NTFS FAT 16 (DnD): 328192-393727]
 - vo18 [Unallocated: 393728-393815]
 - vo19 [NTFS FAT 16 (DnD): 393816-459391]
 - vo23 [Unallocated: 459392-499119]
 - vo24 [NTFS FAT 16 (DnD): 499120-525055]
 - vo27 [Unallocated: 525056-525183]
 - vo28 [NTFS FAT 16 (DnD): 525184-590719]
 - vo11 [Unallocated: 590720-590847]
 - vo32 [NTFS FAT 16 (DnD): 590848-636383]
 - vo35 [Unallocated: 636384-656631]
 - vo36 [NTFS FAT 16 (DnD): 656632-722047]
 - vo39 [Unallocated: 722048-722175]
 - vo40 [NTFS FAT 16 (DnD): 722176-787711]
 - vo43 [Unallocated: 787712-787839]
 - vo44 [NTFS FAT 16 (DnD): 787840-833375]
 - vo47 [Unallocated: 833376-833503]
 - vo48 [NTFS FAT 16 (DnD): 833504-919039]
 - vo49 [Unallocated: 919040-2068479]

File Views

- File Types
 - Images (14)
 - Videos (0)
 - Audio (0)
 - Archives (0)
 - Databases (0)
 - Documents
 - HTML (0)
 - Office (0)
 - PDF (14)
 - Plain Text (0)
 - Rich Text (0)
 - Executable
 - By MIME Type
 - Deleted Files
 - MIME Size
 - Data Artifacts

Listing

Table Thumbnail Summary

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dr)	Flags(Vol)	Known	Location
chartrouse.pdf	1			2022-03-06 21:05:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	37036	Allocated	Allocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo15
crimson.pdf	1			2022-03-06 20:43:44 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	37319	Allocated	Allocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo15
honeyside.pdf	1			2022-03-06 21:20:04 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	37021	Allocated	Allocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo15
whisk.pdf	1			2022-03-06 21:24:56 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	34791	Allocated	Allocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo15
liven.pdf	1			2022-03-06 21:05:40 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	36775	Allocated	Allocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo15
olum.pdf	1			2022-03-06 21:10:50 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	35162	Allocated	Allocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo15
yellow.pdf	1			2022-03-06 20:51:59 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	37647	Allocated	Allocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo15
R188104.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	37319	Unallocated	Unallocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo49
R188256.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	37621	Unallocated	Unallocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo49
R188344.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	34791	Unallocated	Unallocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo49
R188424.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	36775	Unallocated	Unallocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo49
R188536.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	35162	Unallocated	Unallocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo49
R188688.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	37647	Unallocated	Unallocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo49
R188776.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	37636	Unallocated	Unallocated	unknown	Img_InvestigatorDiskImage_aisaid.001/vol_vo49

Pages: 1 of Page Go to Page Script: Latin - Basic

Figure 2: shows a screenshot of a forensic raw copy of the suspect's USB flash drive on the Autopsy program.

Explanation: I used the Autopsy tool to find the name of the color file that found them from searching color code on the WinHex tool to know which file has 8 account numbers.

- what trail of clues you followed?



Figure 3: shows a screenshot of an old coffee-stained scrap of paper with a lot of seemingly random information written on it by a suspect employee found in his office.

Explanation: I have used this paper of evidence to start my investigation which was written by the suspect to find 8 account numbers. I noted that the criminal circle on Partition four with information about tools he used to hide the account number, therefore, I started to analyze a forensic raw copy of the suspect's USB flash drive by searching Partition 4 on the WinHex program to find any information that could help me to find the account number. In addition, I

used the Autopsy tool to search for file color names that figured from its code on winhex by writing it on RGB color mapping stuff website he wrote on hint.

- what your investigation yielded

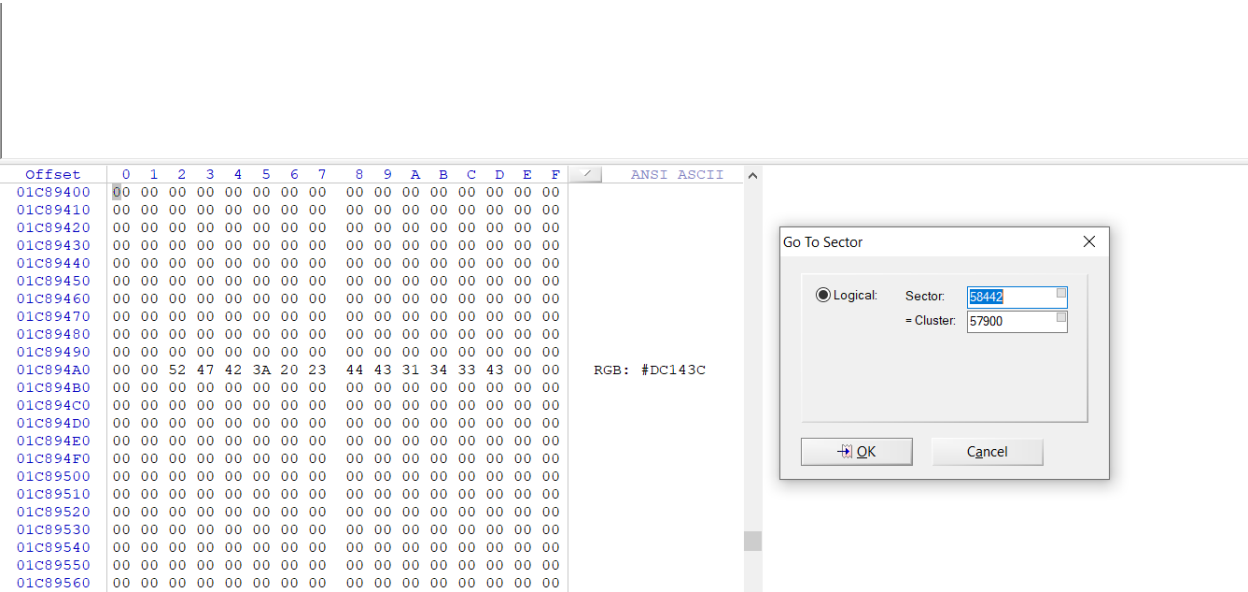


Figure 4: it shows a screenshot of the color code on Partition 14.

Explanation: it shows by clicking on Partition 14 which I discovered from Partition 4 then on Partition 14 went to the left corner by clicking on the sector to write the logical partition sector address (58442) on showing window to provide the color code (RGB :# DC143C) inside the WinHex program under ANSI ASCII.

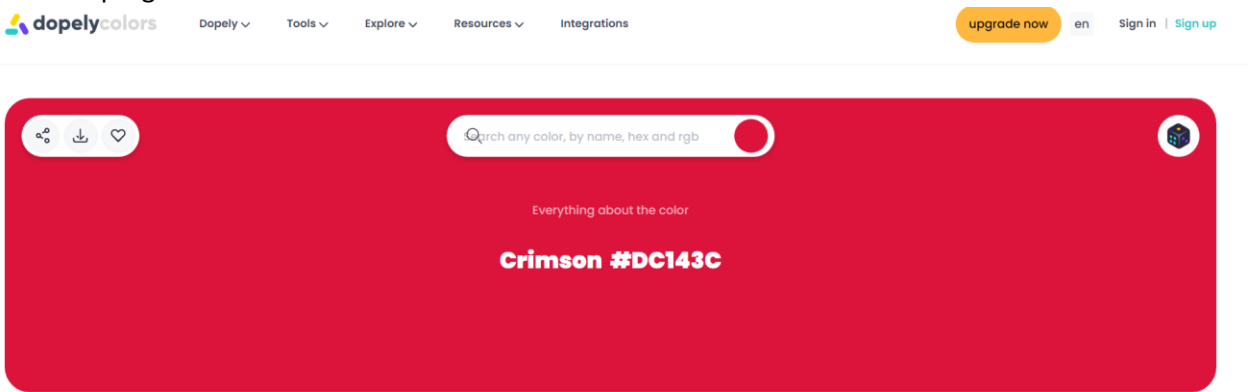


Figure 5: shows a screenshot of the color code that appeared on Partition 14 on the website.

Case - Autopsy 4.19.3
Case View Tools Window Help

Add Data Source Images/Notes Communications Geolocation Timeline Discovery Generate Report Close Case

 Keyword Lists Keyword Search

Data Sources

- InvestigatorClockImage_jailed.001_1heet
- InvestigatorClockImage_jailed.001
- vol1 [Unallocated: 0-127]
- vol2 [RITS / enFAT (bad?) 128-6563]
- vol3 [RITS / enFAT (bad?) 6564-131199]
- vol4 [WinVrS FAT 16 (Dxds) 13200-196730]
- vol7 [Unallocated: 19675-196863]
- vol8 [WinVrS FAT 16 (Dxds) 196864-262390]
- vol11 [Unallocated: 26240-262527]
- vol12 [WinVrS FAT 16 (Dxds) 262528-320863]
- vol13 [Unallocated: 320864-328193]
- vol14 [WinVrS FAT 16 (Dxds) 328192-393727]
- vol15 [Unallocated: 393728-393850]
- vol20 [WinVrS FAT 16 (Dxds) 393856-499220]
- vol23 [Unallocated: 49920-499393]
- vol24 [WinVrS FAT 16 (Dxds) 499320-525050]
- vol27 [Unallocated: 525056-525183]
- vol28 [WinVrS FAT 16 (Dxds) 525184-590718]
- vol31 [Unallocated: 590720-590847]
- vol32 [WinVrS FAT 16 (Dxds) 590848-656383]
- vol35 [Unallocated: 656384-656511]
- vol36 [WinVrS FAT 16 (Dxds) 656512-722047]
- vol39 [Unallocated: 722048-722175]
- vol40 [WinVrS FAT 16 (Dxds) 722176-787111]
- vol43 [Unallocated: 787112-787830]
- vol44 [WinVrS FAT 16 (Dxds) 787840-853370]
- vol47 [Unallocated: 853376-853503]
- vol48 [WinVrS FAT 16 (Dxds) 853504-919030]
- vol49 [Unallocated: 919040-2068470]

Listing PDF Table Thumbnail Summary

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flag(Dr)	Flag(Wr)	Known	Location
✓ charmsrun.pdf	1			2022-03-06 21:05:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	37836	Allocated	Allocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ charmsrun.pdf	1			2022-03-06 20:53:44 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	37310	Allocated	Allocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ honeydave.pdf	1			2022-03-06 21:20:04 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	37021	Allocated	Allocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ lthak.pdf	1			2022-03-06 21:24:56 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	34791	Allocated	Allocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ ltrun.pdf	1			2022-03-06 21:15:40 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	36775	Allocated	Allocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ ulum.pdf	1			2022-03-06 21:10:50 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	35102	Allocated	Allocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ yllow.pdf	1			2022-03-06 20:51:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	2022-03-06 21:46:58 CST	37647	Allocated	Allocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ 1188164.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	37313	Unallocated	Unallocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ 1188236.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	37021	Unallocated	Unallocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ 1188344.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	34791	Unallocated	Unallocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ 1188424.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	36775	Unallocated	Unallocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ 1188536.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	35102	Unallocated	Unallocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ 1188684.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	37647	Unallocated	Unallocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1
✓ 1188875.pdf	1			0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	37836	Unallocated	Unallocated	unknown	Img_InvestigatorClockImage_jailed.001\vol1

File Views

- By Extension
- Images (14)
- Notes (0)

Explanation: I opened a forensic raw copy of the suspect's USB flash drive on the Autopsy program to look for anything related to the crimson color I found a PDF file has this name.

- 8 account numbers found inside of crimson file

97084583

Figure 7: it shows a screenshot containing the crimson file which shows 8 account numbers.

Explanation: after I extracted the file, I could not open it, so I open the file on WinHex to clean file by searching the type of extension file PDF under ANSI ASCII to fix it by changing the header signature which I found on (List of file signatures - Wikipedia [List of file signatures - Wikipedia](#)) to open it.

- The false clue but I open it because I found on Partition 4 which is part of answer

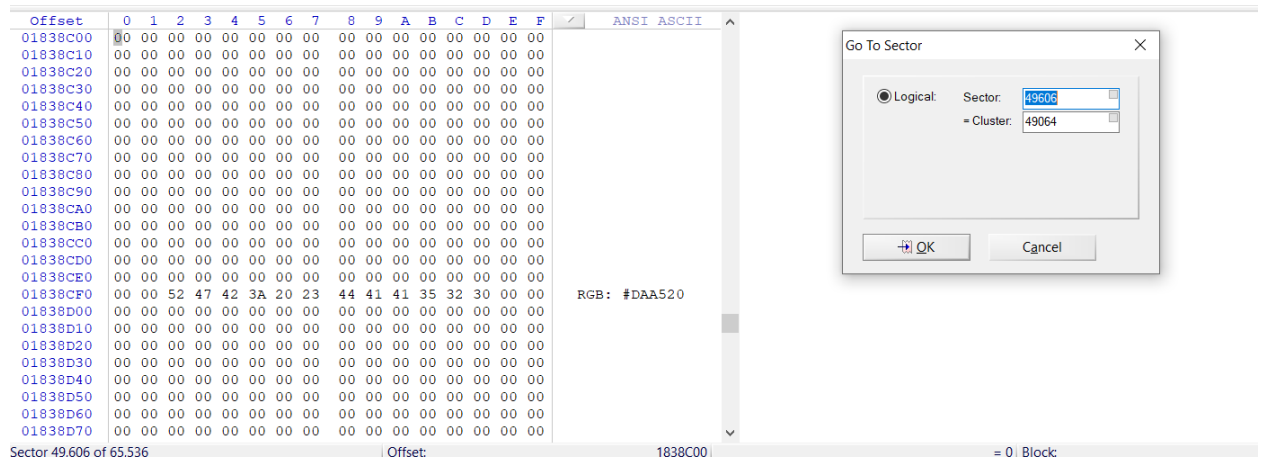


Figure 8: shows a screenshot of the color code on Partition 12.

Explanation: it shows by clicking on Partition 12 which I discovered from Partition 4 then on Partition 12 went to the left corner by clicking on the sector to write the logical partition sector address (49606) on showing window to provide the color code (RGB :# DAA520) inside the WinHex program under ANSI ASCII.

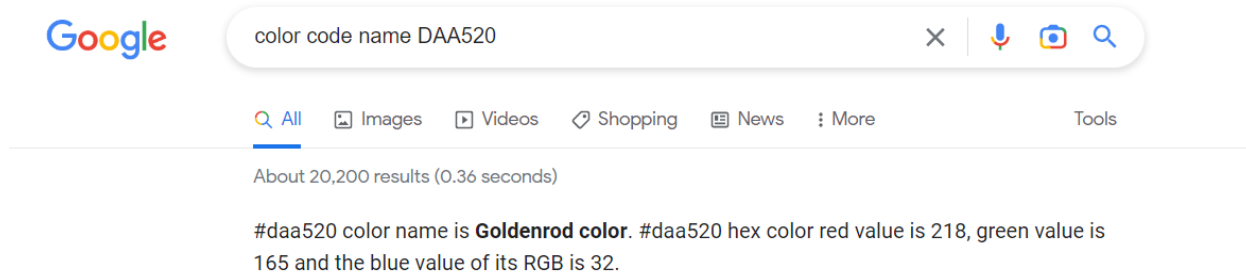


Figure 9: it shows a screenshot of the color code that appeared on Partition 12 on google.

Explanation: after I found the color code, I looked to hint paper to see color map stuff. As a result, I searched on google by writing the code color that I found on WinHex to provide me the name of the color which is Goldenrod Color.

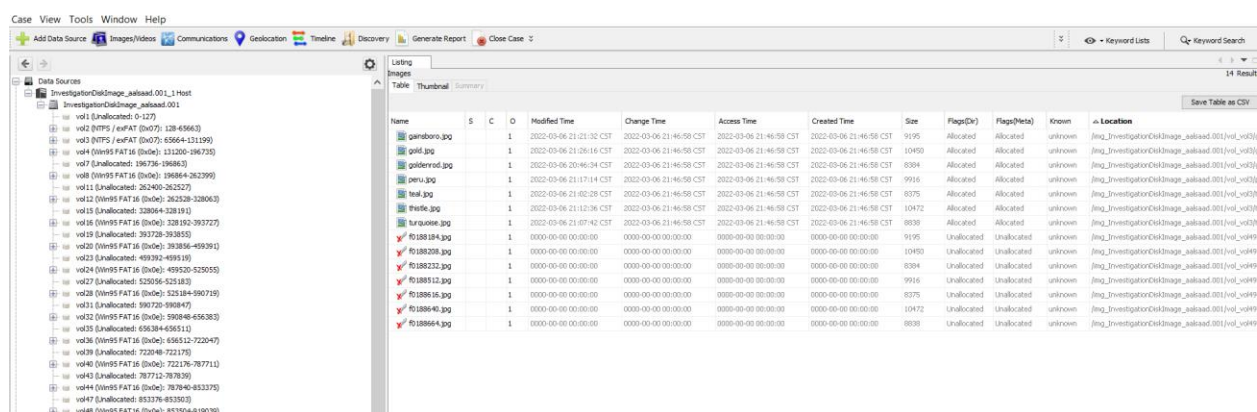


Figure 10: shows a screenshot of the Goldenrod file found on the Autopsy tool.

Explanation: I opened a forensic raw copy of the suspect's USB flash drive on the Autopsy program to look for anything related to Goldenrod color I found a jpg file that has this name.

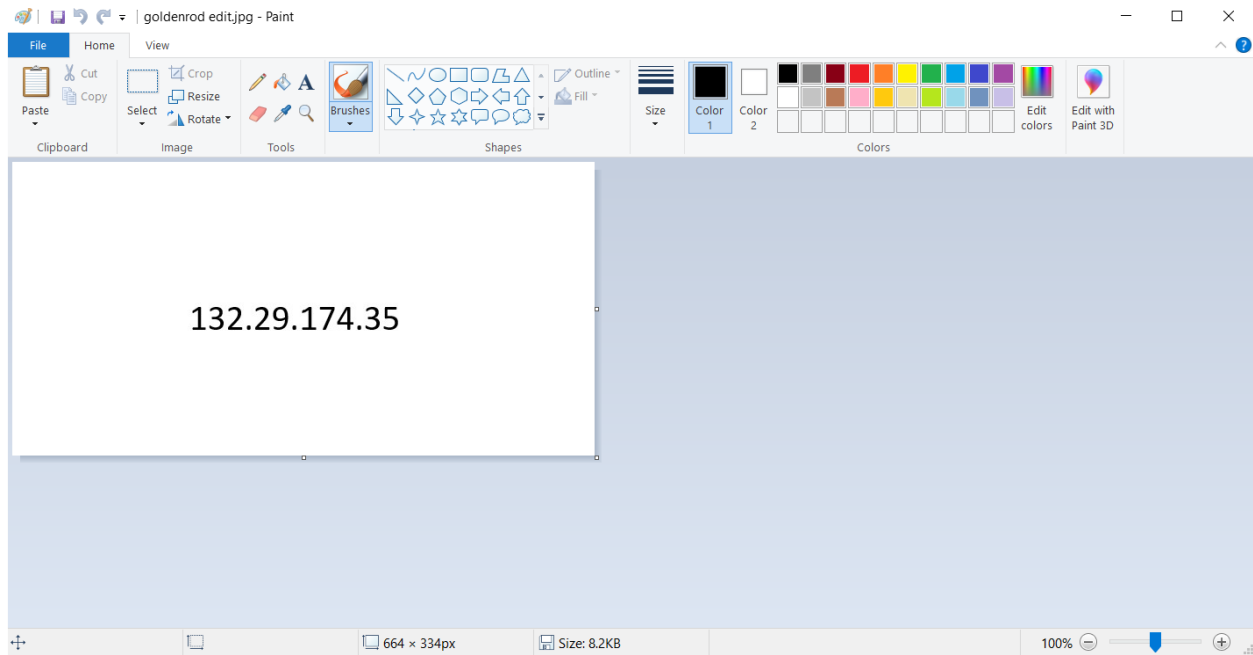


Figure 11: it shows a screenshot containing the Goldenrod file which shows a false clue IP address opened by the paint program.

Explanation: After I extracted the file, I could not open it, so I open the file on WinHex to clean file by searching the type of extension file JPG under ANSI ASCII to fix it by changing the header signature which I found on (List of file signatures - Wikipedia [List of file signatures - Wikipedia](#)) to open it.